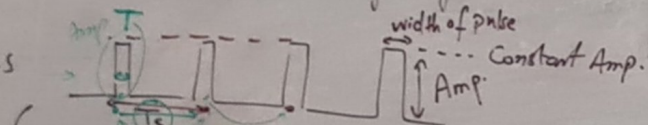


Pulse Modulation.

- refer to discretization of analog signal.

→ used train of pulses



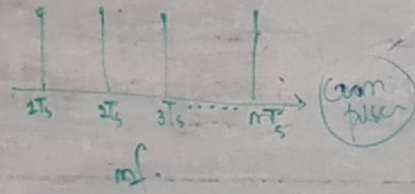
→ this convert information signal into discrete signal.

$T_s \rightarrow$ Sampling time.

$T \rightarrow$ width of pulse.

→ First step to digital system is to convert analog signal into discrete signal.

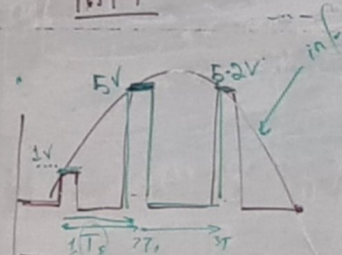
pulse Mod.



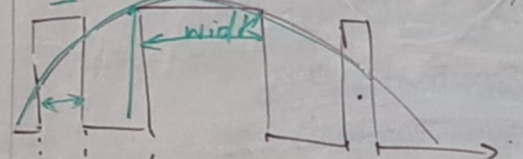
② ③ ④

Why we need digital comm.?

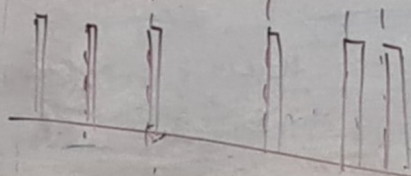
PAM



PWM



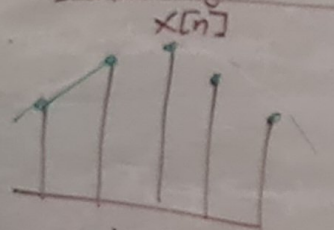
PPM



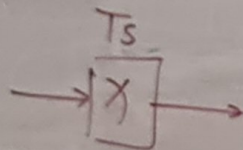
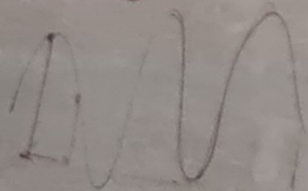


Pulse Modulation.

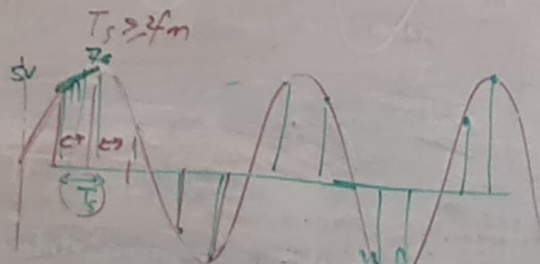
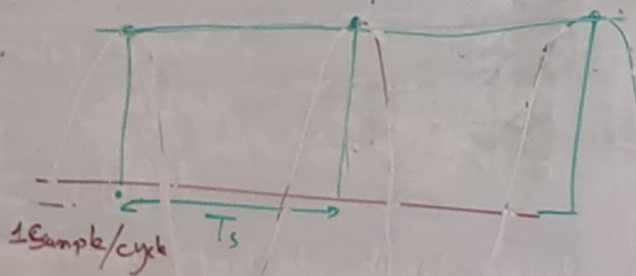
Sampling Theorem.



amplitude could be at any value but time is discrete



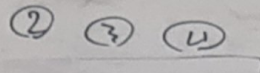
Instantaneous Sampling.



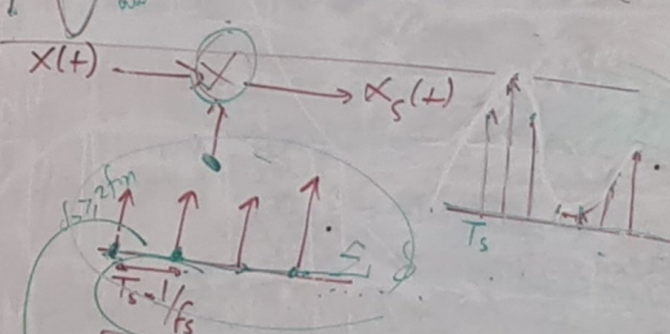
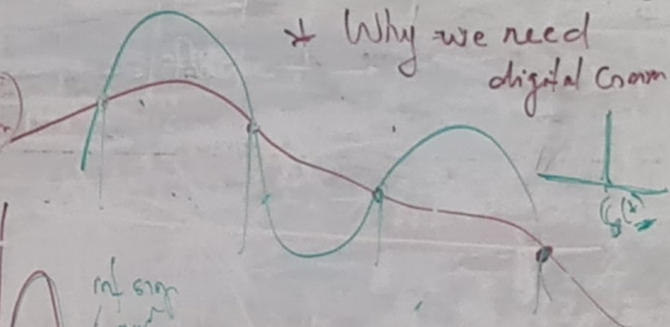
5 sample/cycle

$$T_s = \frac{1}{f_s}$$

$$f_s \geq 2f_m$$



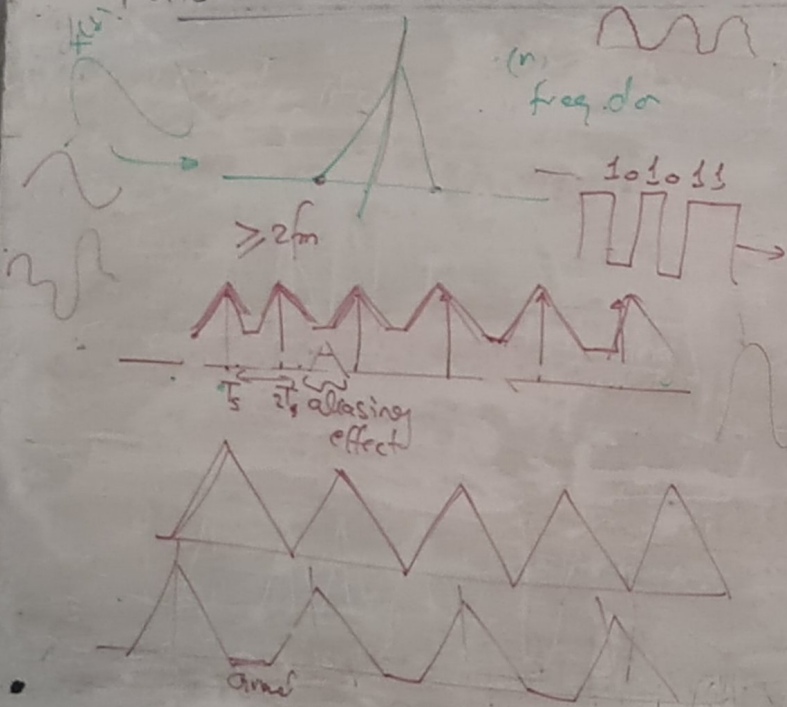
Why we need digital comm.?



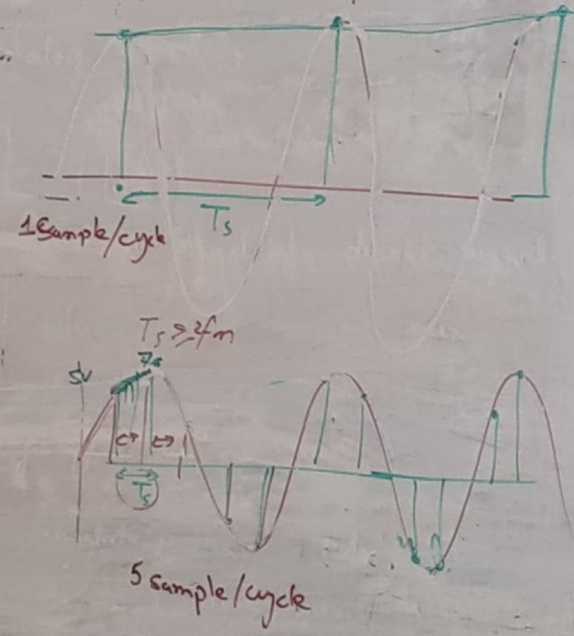
$$x(t) \cdot \sum \delta(t - nT_s) = x_s(t)$$

Nyquist rate $f_s \geq 2f_m$

Pulse Modulation



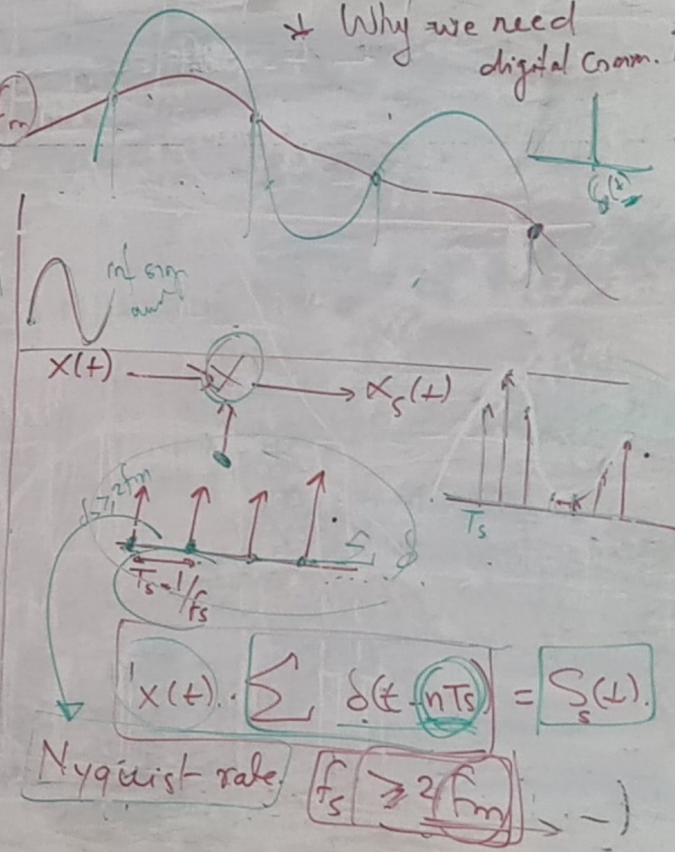
Instantaneous Sampling



$$T_s = \frac{1}{f_s}$$

$$f_s \geq 2f_m$$

Why we need digital comm.?



$$x(t) \cdot \sum \delta(t - nT_s) = x_s(t)$$

$$f_s \geq 2f_m$$

$$\frac{1}{f_s} = \text{Nyquist interval}$$

$$x(t) = 3\cos(50\pi t) + 10\sin(300\pi t) + \cos(100\pi t)$$

Find Nyquist rate. Nyquist interval.

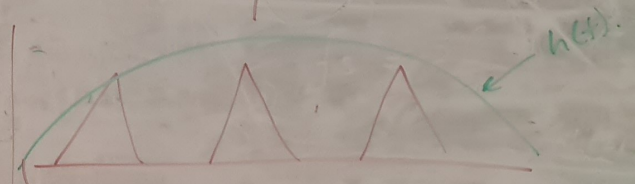
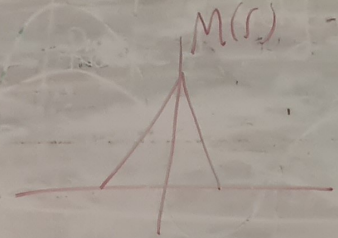
$$f_s \geq 2f_{\max}$$

$$\frac{1}{f_s} \rightarrow \text{Nyq. Int.}$$

Pulse Modulation

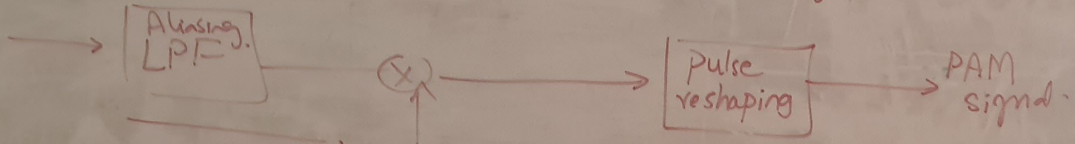
FLAT Top PAM

conventional PAM



to remove it we used (LPF).

$$m_s(t) = \sum m(nT_s) \cdot \delta(t - nT_s)$$



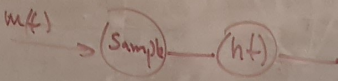
basic form of pulse generation
each pulse has a period of T.

pulse generator where
 $f_s \geq 2f_m$

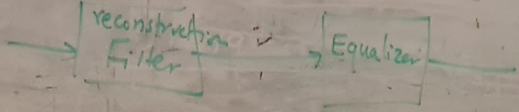
$\frac{T}{T_s} \approx 0.1$ as T increases distortion ↑.

$$S(t) = m_s(t) * h(t) \rightarrow \text{in frequency domain: } M(f) \text{ LPF } (T \text{ sinc}(fT)) e^{-j\pi fT}$$

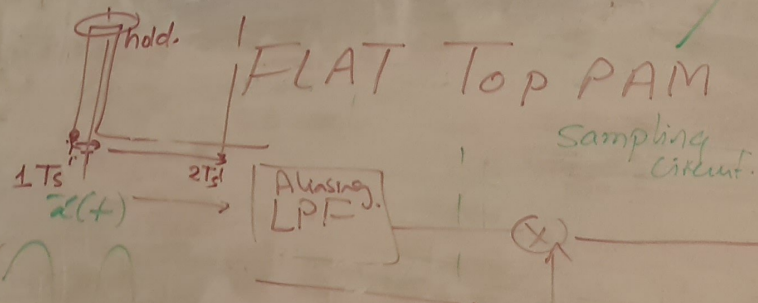
amplitude distortion → delay in time



we remove it using $\frac{1}{T \text{ sinc}(fT)}$



Pulse Modulation

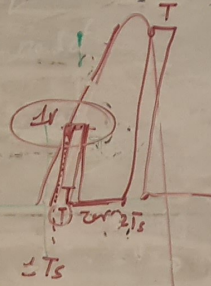


basic form of pulse generator
each pulse has a period of T .

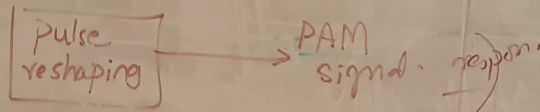
$$\frac{T}{T_s} < 0.1 \quad \text{as } T \uparrow \text{ increase distortion } \uparrow$$

pulse generator where
 $f_s \geq 2f_m$

conventional PAM

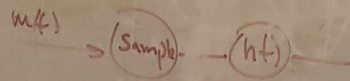


$$m_s(t) = \sum m(nT_s) \cdot \delta(t - nT_s)$$



$$S(t) = m_s(t) * h(t)$$

$$h(t) = \begin{cases} 1, & \text{if } 0 < t < T \\ 0, & \text{other wise.} \end{cases}$$

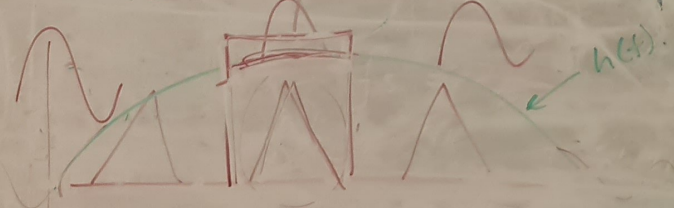
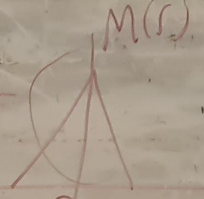
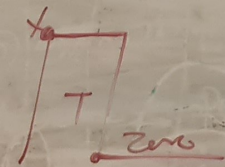
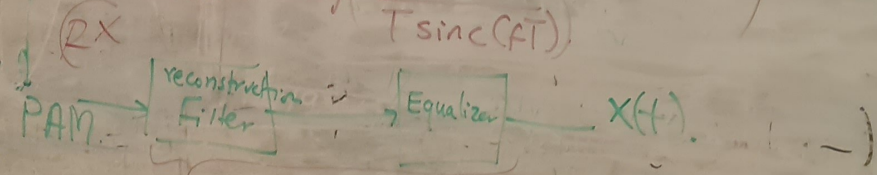


in frequency domain:

$$M(f) \left(\text{LPF } T \text{sinc}(fT) e^{-j\pi fT} \right)$$

amplitude distortion, delay in time, delay

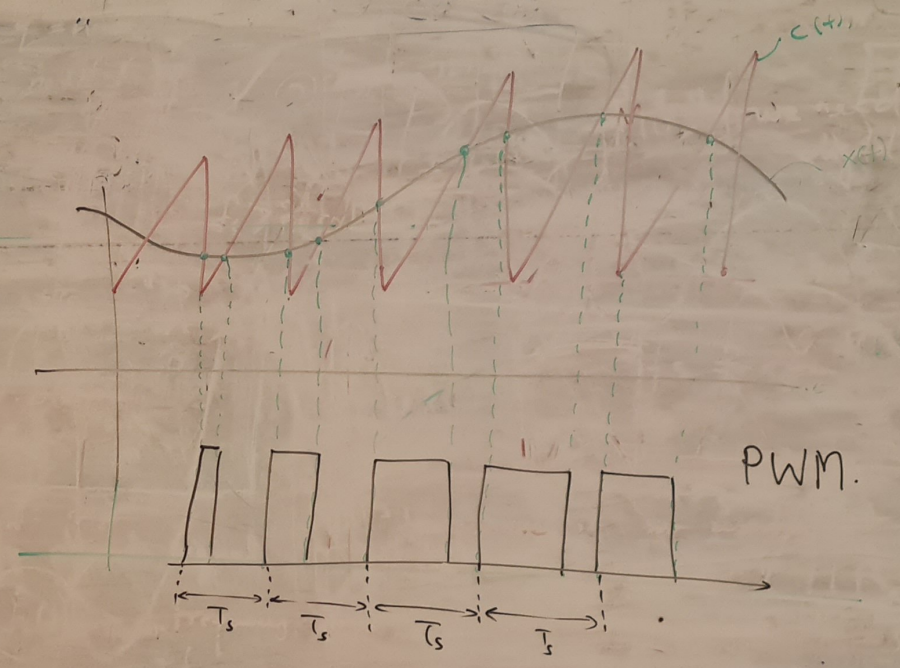
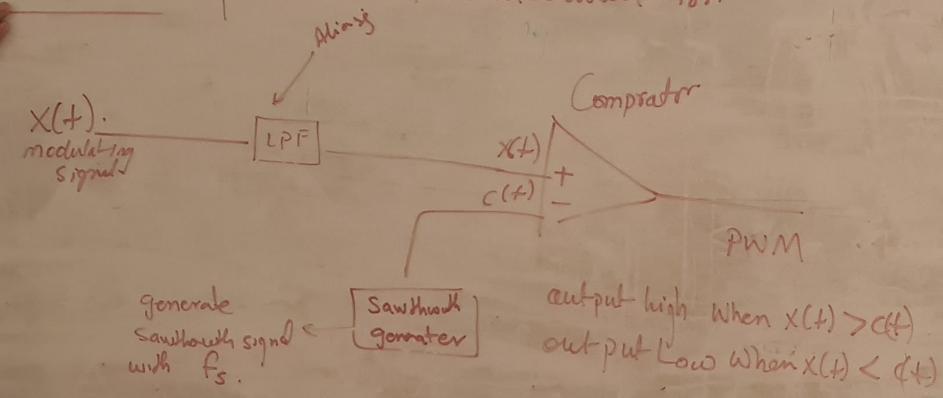
we remove it using $\frac{1}{T \text{sinc}(fT)}$



to remove it we used (LPF).

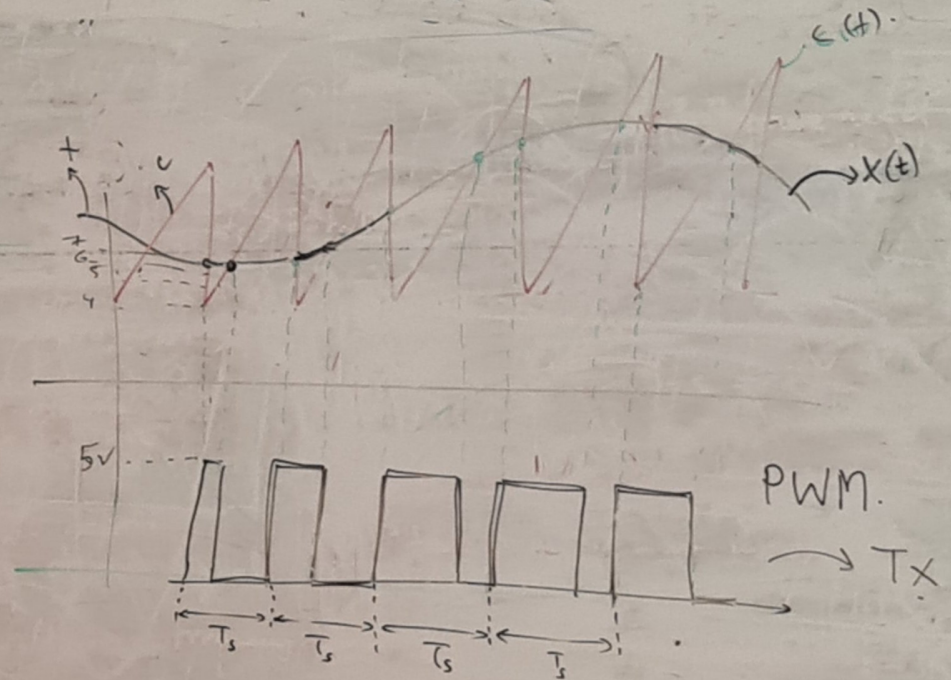
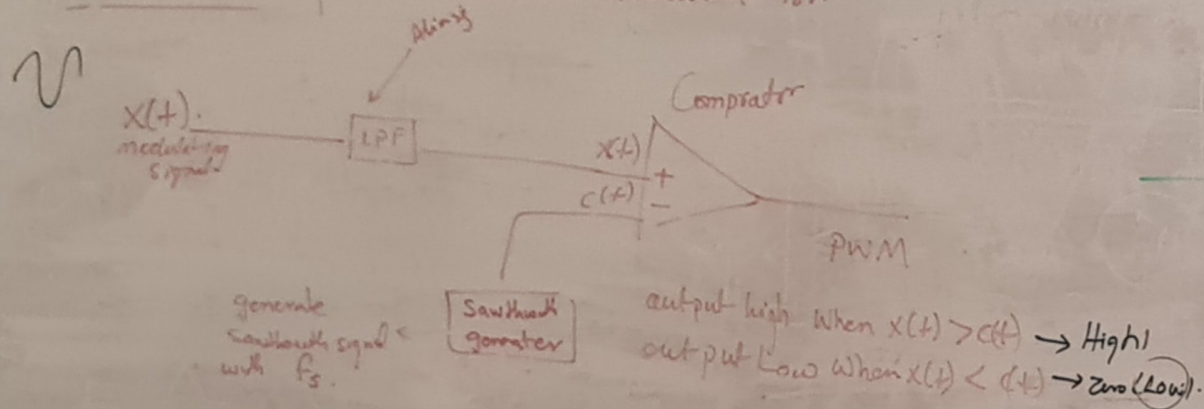
Pulse Modulation

PWM. pulse width modulation.



Pulse Modulation

PWM. pulse width modulation.

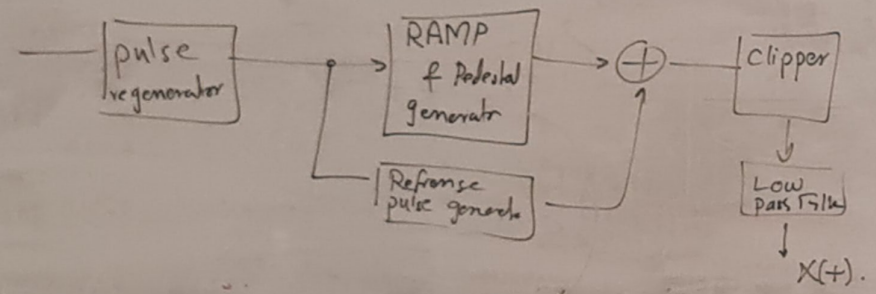


① ②

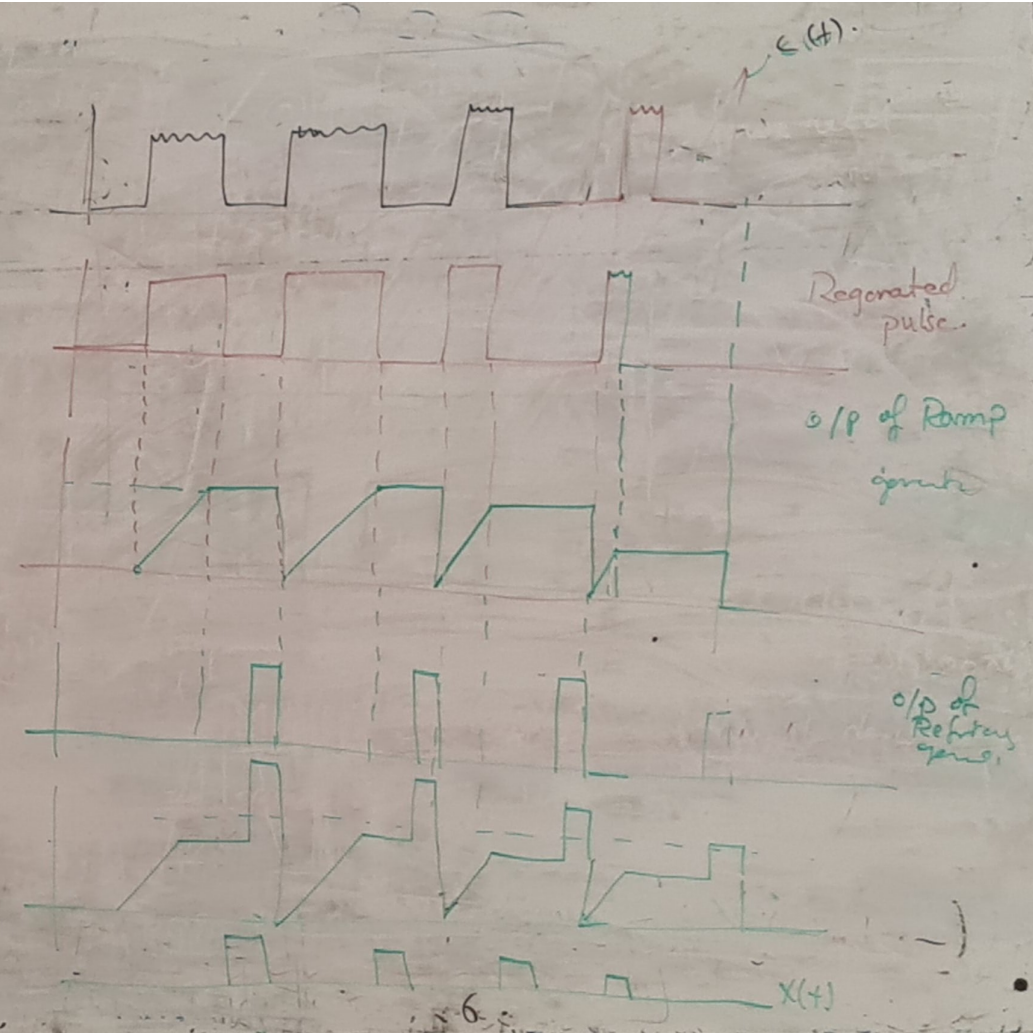
Pulse Modulation

PWM. pulse width modulation.

Detection of PWM.



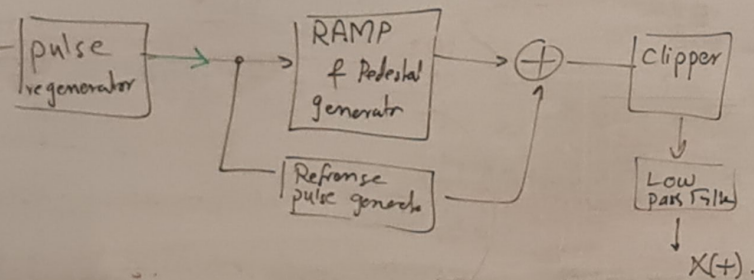
Slope or ramp = duration of pulse.



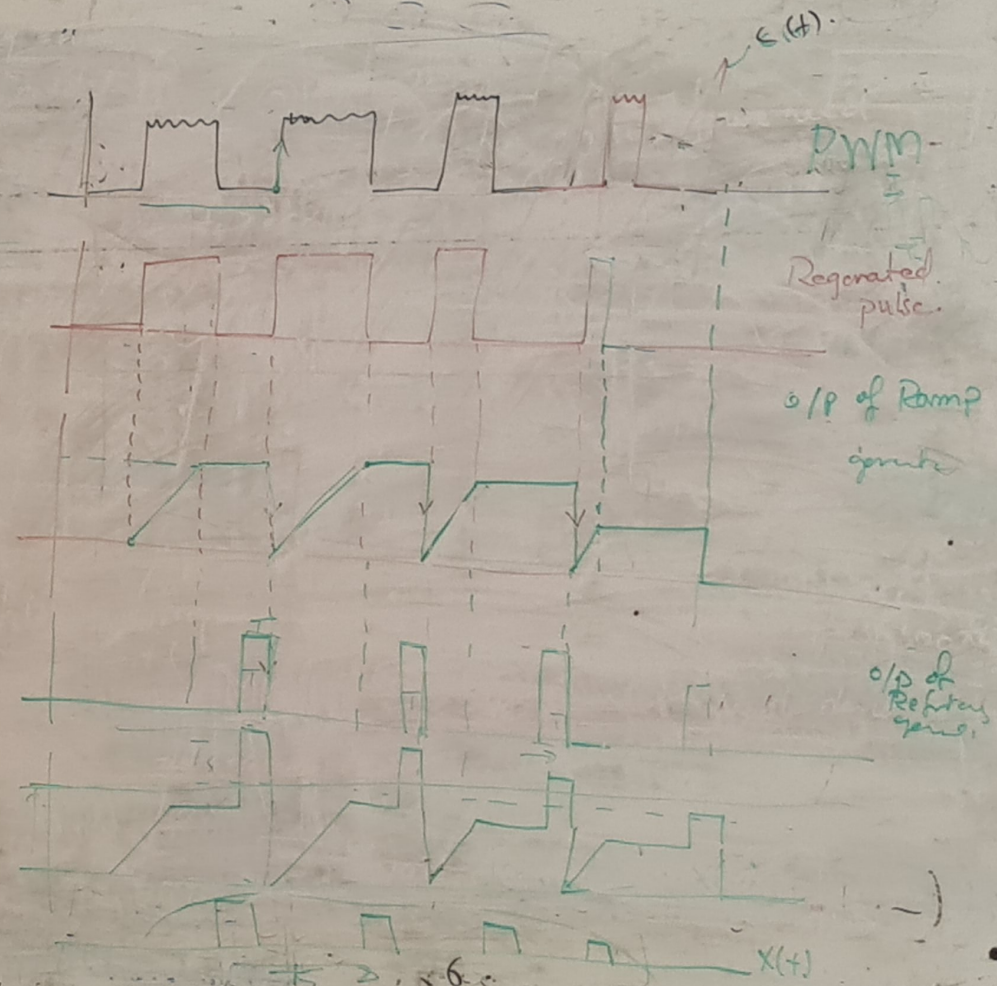
Pulse Modulation

PWM. pulse width modulation.

Ex PWM Detection of PWM.



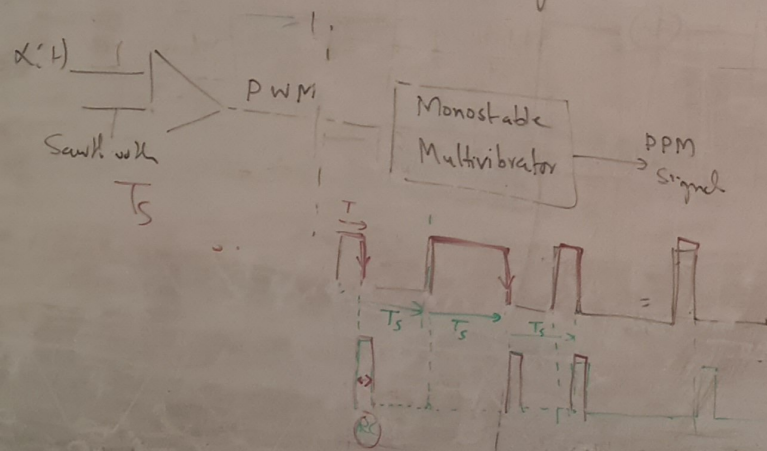
Slope or ramp = duration of pulse.



Pulse Modulation

Pulse Position Mod.

Generation of PPM signal.



demodulation

